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CSE-C

DAY -6

SALESFORCE TRAINING PROGRAM

Day 6 – Making Software Talk to Data (SOQL & DML)

Objective

The objective of this task was to understand how Salesforce retrieves and stores data using SOQL and DML. In this practical, I worked on the Placement Management System project and implemented the application submission process. The system retrieves student and job information, checks whether the student is eligible, prevents duplicate applications, and finally creates a new application record.

Retrieving Student Information

The first step was to retrieve the student details from the **Student** object. I used a SOQL query to fetch the student's CGPA based on the Student Id. This information is required to verify whether the student is eligible for the selected job.

```
Student__c student = [  
    SELECT CGPA__c  
    FROM Student__c  
    WHERE Id = :studentId  
    LIMIT 1  
];
```

After executing the query, I verified that the student's CGPA was successfully retrieved from Salesforce.

Retrieving Job Information

After retrieving the student details, I fetched the job information from the **Job** object. The main purpose of this query was to retrieve the minimum CGPA required for the selected job.

```
Job__c job = [  
    SELECT Minimum_CGPA__c  
    FROM Job__c  
    WHERE Id = :jobId  
    LIMIT 1  
];
```

Instead of using a fixed CGPA value, the program retrieves the required CGPA directly from the job record, making the application process more flexible.

Screenshot: *(Insert screenshot of the Job SOQL query.)*

Checking for Duplicate Applications

Before creating a new application, I added a duplicate check. This prevents a student from applying for the same job more than once.

```
List<Application__c> existingApplications = [  
    SELECT Id  
    FROM Application__c  
    WHERE Student__c = :studentId  
    AND Job__c = :jobId  
    LIMIT 1  
];
```

If an existing record is found, the system returns a message informing the user that they have already applied for that job.

Screenshot: *(Insert screenshot of the duplicate check query.)*

Eligibility Validation

Once the student and job information was retrieved, I compared the student's CGPA with the minimum CGPA required for the job.

```
if(student.CGPA__c < job.Minimum_CGPA__c){  
    return 'You are not eligible.';
```

```
}
```

This validation ensures that only eligible students can submit their applications.

During testing, I initially selected a student whose CGPA was lower than the required value. The system correctly prevented the application from being submitted. After selecting another student with sufficient CGPA, the application was submitted successfully.

Screenshot: *(Insert screenshot of the eligibility validation code.)*

Creating the Application Record

After all the validations were completed successfully, I created a new Application record and assigned the required field values.

```
Application__c application = new Application__c();
```

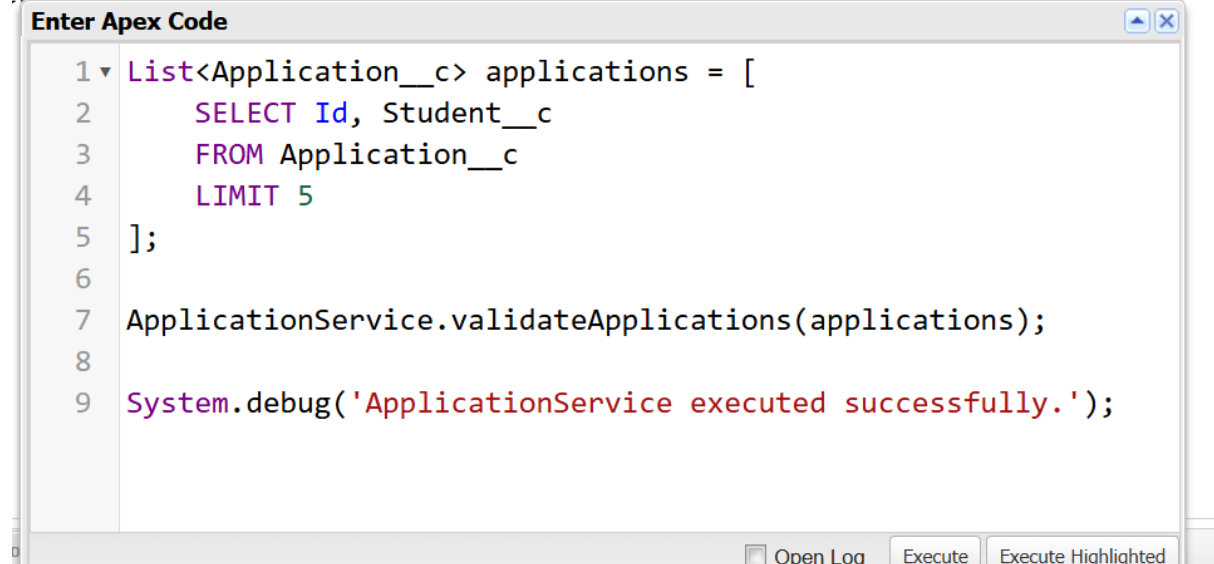
```
application.Student__c = studentId;
```

```
application.Job__c = jobId;
```

```
application.Application_Date__c = Date.today();
```

This object stores the details of the student's job application.

Service.validateApplications(List<Application__c> applications) {



Saving the Record Using DML

To store the application in Salesforce, I used the DML **insert** statement.

```
insert application;
```

After executing this statement, a new Application record was successfully created in Salesforce.



Updating the Application Status

I also implemented a method to update the status of an existing application. This method retrieves the application record, changes its status, and updates it in Salesforce.

```
application.Status__c = newStatus;
```

```
update application;
```

While testing this functionality, I encountered a restricted picklist error. After verifying the object configuration and testing with valid data, I understood that Salesforce validates picklist values before updating the record.



Testing the Application

I tested the application using the **Execute Anonymous Window** in the Developer Console.

```
String result = ApplicationService.submitApplication(studentId, jobId);
```

```
System.debug(result);
```

Initially, I tested with a student who did not satisfy the eligibility criteria, and the system correctly prevented the application. Later, I tested with another eligible student, and the application was submitted successfully. A new Application record was created, and a confirmation email was received.

Execution Log		
Timestamp	Event	Details
11:44:45:046	USER_DEBUG	[9] [[DEBUG] (Application__c: {Id=a09dm00000UzxOTQAZ, Student__c=a07dm000001ZpEvaQAF, Status__c=Eligible}, Application__c: {Id=a09dm00000UzxyRQAR, Student__c=a07dm000001ZodqoQAB, Status__c=Eligible}, Application__c: {Id=a09

Through this practical, I understood the importance of SOQL and DML in Salesforce development. I learned how to retrieve data using SOQL, apply business validations, prevent duplicate records, create new records using DML, and test Apex methods using the Execute Anonymous Window. This practical also helped me understand how different Salesforce components work together to complete a real business process.

In this task, I successfully implemented the application submission process for the Placement Management System. The system retrieves student and job details, validates eligibility, prevents duplicate applications, and creates a new application record only when all business conditions are satisfied. After testing the implementation with different student records.

GITHUB: [LINK](#)